



CRYPTOLOGY

B15



WHAT IS IT?



The study and analysis of cryptographic techniques.

In a nutshell 🥜:

It's the study of how we secure digital communications



2 BRANCHES OF CRYPTOLOGY

CRYPTOGRAPHY

Creating the secure text

✦ Turning text that can be read by humans into a unreadable text - unless you have the 'secret' key

CRYPTOANALYSIS

Breaking the hidden message

✦ The study of trying to crack the system that converts the text to unreadable, without having the 'secret' key

HOW DATA IS SECURED – ASYMMETRICAL KEY ENCRYPTION

PUBLIC KEY

PRIVATE KEY

Everyone has both public and private keys.

Known to everyone.

Only known to you

If you send a message you encrypt the message with the receivers public key.

To decrypt the key back to readable text, you must own the corresponding private key.

WHAT THIS MEANS

Ciphertext

5fcfd41e547a12215b1

VS

Plaintext

trustno1

Asymmetric encryption

